Yihang She

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Education

University of Cambridge 2022–2026
PhD in Computer Science; expected submission: November 2026
UKRI Centre for Doctoral Training in Artificial Intelligence for Environmental Risks
Supervisors: Srinivasan Keshav and Andrew Blake

ETH Zurich 2019–2022
MSc in Geomatics; GPA: 5.3/6.0. Excellence Scholar.
Scholarship mentor: Konrad Schindler

Nanjing University 2015–2019
BSc in Geographic Information Science; GPA: 4.6/5.0. Graduated with distinction.
Exchange student, University of California, Berkeley, Spring 2018; GPA: 3.9/4.0

Research Themes

AI for scientific discovery, computer vision, physics-informed machine learning, synthetic data, and remote sensing

Manuscripts and Peer-Reviewed Publications

[1] **Yihang She**, Andrew Blake, and Srinivasan Keshav. "PiCo: Physics-Informed Concept Probing for Geo-Foundation Models." Manuscript prepared for submission to *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

[2] **Yihang She**, Andrew Blake, Clement Atzberger, Adriano Gualandi, and Srinivasan Keshav. "PILA: Physics-Informed Low Rank Augmentation for Interpretable Earth Observation." Accepted for publication conditioned on minor revisions, *Journal of Machine Learning Research*.

[3] **Yihang She**, Andrew Blake, David Coomes, and Srinivasan Keshav. "Scaling Up Forest Vision with Synthetic Data." *International Journal of Computer Vision*, 2026.

[4] Zhengpeng Feng, **Yihang She**, and Srinivasan Keshav. "SPREAD: A Large-Scale, High-Fidelity Synthetic Dataset for Multiple Forest Vision Tasks." *Ecological Informatics*, 2025.

[5] **Yihang She**, Goutam Bhat, Martin Danelljan, and Fisher Yu. "Fast Hierarchical Learning for Few-Shot Object Detection." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.

[6] Riccardo de Lutio, **Yihang She**, Stefano D'Aronco, Stefania Russo, Phillipp Brun, Jan Dirk Wegner, and Konrad Schindler. "Digital Taxonomist: Identifying Plant Species in Citizen Scientists' Photographs." *ISPRS Journal of Photogrammetry and Remote Sensing*, 2021.

[7] **Yihang She***, Zihan Liu*, Wenfeng Zhan, Jiameng Lai, and Fan Huang. "Strong Regulation of Daily Variations in Nighttime Surface Urban Heat Islands by Meteorological Variables across Global Cities." *Environmental Research Letters*, 2021. *Equal contribution.

Honours, Fellowships, and Awards

Cambridge Centre for Carbon Credits PhD Fellowship 2022–2026
Fully funded four-year PhD studentship awarded to international students

Excellence Scholarship and Opportunity Programme, ETH Zurich 2019–2021
Fully funded master's scholarship awarded to the top 3% of students in each cohort

Outstanding Graduate, Nanjing University 2019
Awarded to students in the top 5% of the graduating cohort based on academic performance

National Scholarship, Ministry of Education of China 2016
Awarded to the top 0.2% of students nationwide based on academic excellence

Invited Talks and Media

Professor Bill Freeman's Group, MIT, USA October 2026
Invited talk, *Bridging Physics and AI for Interpretable Earth Observation*

Intelligent Earth Conference, Oxford, UK October 2026
Invited talk, *Scaling Up Forest Vision with Synthetic Data*

Meet the Researcher Teaching AI Drones to Count Trees July 2026
Research highlight on the Department of Computer Science and Technology website, Cambridge

Cambridge Centre for Climate Science Symposium, University of Cambridge, UK April 2026
Invited talk, *PILA for Interpretable Earth Observation*

3DGeo Research Group Seminar, Heidelberg University, Germany October 2025
Invited talk, *Cambridge Arboreal Modelling Panoptic 3D (CAMP3D) Simulation Toolkit*

PBS Documentary: *The Future of Nature* March 2025
Rendered LiDAR point clouds from my work on the Scottish Cairngorms were featured

Presentations at Conferences and Workshops

The UChicago and Caltech AI+Science Conference, Chicago, USA October 2026
PILA accecpted for a poster presentation.

Conference on Computer Vision and Pattern Recognition, Denver, USA June 2026
Poster presentations at the Synthetic Data for Computer Vision Workshop and the Computer Vision for Science Workshop.

International Computer Vision Summer School, Sicily, Italy July 2025
Poster presentation on "Scaling Up Forest Vision with Synthetic Data".

EurIPS, Copenhagen, Denmark December 2025
Poster presentation at the Representation Learning for Earth Observation Workshop.

International Computer Vision Summer School, Sicily, Italy July 2024
Poster presentation on physics-informed learning for remote sensing and geophysics.

International Conference on Learning Representations, Vienna, Austria May 2024
Poster presentation at the Climate Change AI Workshop.

ELLIS Computer Vision Workshop, Modena, Italy April 2024
Poster presentation on "From Spectra to Biophysical Insights: End-to-End Learning with a Biased Radiative Transfer Model."

IEEE/RSJ International Conference on Intelligent Robots and Systems, Kyoto, Japan October 2022
Oral presentation on 'Fast Hierarchical Learning for Few-Shot Object Detection."

Selected Training and Work Experience

EnterpriseTECH Cohort 16, Cambridge Judge Business School, University of Cambridge 2024–2025
Selected for an entrepreneurship programme for tech-innovation and worked on a biotechnology project under the supervision of Vincenzo Garzya, Head of Data and AI at AstraZeneca.

Research Intern, Elevate3D, Zurich 2022
Adapted segmentation models for NeRF-based 3D reconstruction and developed a Blender pipeline in Python for generating synthetic 3D-object data.

Research Assistant, ETH Zurich 2022
Extended a crop-mapping method to nationwide coverage across Switzerland and developed a data processing pipeline from raw satellite observations to analysis-ready data.

Project Lead, Computational Intelligence Lab taught by Thomas Hofmann, ETH Zurich 2020
Developed a U-Net-based system for road segmentation in satellite imagery.

Project Lead, 3D Vision Lab taught by Marc Pollefeys, ETH Zurich 2020
Developed an augmented-reality language-learning application using Microsoft HoloLens.

Mitacs Globalink Research Intern, University of Winnipeg 2018
Identified research gaps in community forestry in Canada for environmental governance.

Technical Skills

Data modalities and sensors: Experience with data in both 2D and 3D, including RGB, multispectral, LiDAR, and geodetic data from platforms ranging from drones to satellites.

3D vision and scientific machine learning: Experience with Unreal Engine, Blender, and graphics simulators; developed the pip-installable CAMP3D synthetic-data generation toolkit and implemented a 2,000-line radiative transfer model in PyTorch.

Research adaptability: Demonstrated ability to rapidly develop expertise and publish across leading venues. Research spans 3D vision, physics-informed machine learning, and explainable AI within a coherent framework.

Languages: Chinese (native), English (fluent), Korean (basic), and German (basic).

Teaching and Demonstration

Scientific Demonstrator, Cambridge Festival, University of Cambridge 2024, 2026

Tutor, Undergraduate Course in Further Graphics, University of Cambridge 2024, 2025, 2026

Scientific Demonstrator, Department of Computer Science and Technology Undergraduate Open Days, University of Cambridge 2024

Tutor, Undergraduate Course in Artificial Intelligence, University of Cambridge 2024

Scientific Demonstrator, Cambridge Zero Community Day, University of Cambridge 2024

Hobbies

Choral singing (tenor, Chapel Schola): recipient of the 2023 Choral Exhibition Award and the 2024 Instrumental Award from St Edmund's College, Cambridge

Chinese calligraphy: more than 20 years of practice since childhood

Other interests: hiking, skiing, and reading